

How strongly do green spaces in neighbourhoods affect Airbnb offer and demand in Melbourne?

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Abstract

This project looks at how parks and trees affect Airbnb supply and demand in Melbourne. By combining Airbnb data with the city’s 2025 tree inventory, we analyzed over 24,000 listings and reviews using text-mining and mapping tools. We found that reviews mentioning green spaces are significantly more positive, and listings that use nature-related language attract 93% more reviews on average. Over 86% of hosts use green keywords to market their properties, with these listings heavily concentrated in park-rich and coastal neighborhoods like Port Phillip and Yarra. Our analysis shows that guests strongly link nature terms with positive feelings like comfort and quiet. These findings show that urban greenery heavily shapes the local short-term rental market, adding important evidence to ongoing debates about housing availability and who benefits from public investments in city parks.

1 Introduction

1.1 Context

Airbnb has grown into one of the most popular ways to find short-term accommodation, and Melbourne is no exception. With thousands of listings spread across the city, guests leave behind large amounts of written reviews that capture what they thought about their stay. These reviews mention everything from how clean the apartment was to whether there was a park nearby or a nice view out the window. The question we wanted to explore is whether that kind of language, specifically mentions of green spaces like parks, gardens, trees, and nature trails, shows up more often in positive reviews and whether listings in greener areas tend to charge more.

1.2 Policy and Social Debates

Melbourne sits within a regulatory environment that has become increasingly restrictive for short-term rental platforms. Victoria introduced new rules that require hosts to register their property with the state government and pay a seven and a half percent levy on short-stay accommodation revenue. The City

of Melbourne and several inner-suburban councils have also pushed for tighter controls, particularly in residential areas where long-term tenants and neighbours have raised concerns about noise, building wear, and housing availability.

These rules matter for our analysis for a couple of reasons. First, the distribution of listings is not random. Certain neighbourhoods, especially those close to the CBD, the Yarra River, and parks like Fitzroy Gardens or Royal Park, tend to have higher concentrations of active listings. If green space is both a desirable amenity and a predictor of listing density, then any premium we find could reflect location desirability more broadly rather than trees and gardens specifically. Second, the levy and registration requirements have pushed some hosts out of the market, which means the listings we observe are likely skewed towards more professionally managed properties, not casual hosts renting out a spare room.

There is also a broader housing debate in Melbourne worth noting. Critics of platforms like Airbnb argue that converting residential properties into short-term rentals reduces the supply of long-term rental housing, which drives up rents for ordinary residents. From that angle, a finding that green neighbourhoods drives

Airbnb demand can question who benefits from urban green infrastructure. In other words, we would see whether wealthier, leafier suburbs are disproportionately converted into tourist accommodation.

1.3 Data

To investigate this, a dataset was used from Inside Airbnb, which provides publicly available information on Melbourne listings, reviews, calendar availability, and neighbourhood boundaries. In addition, we crossed it with the data from Melbourne City open data website. We created a map of every tree in the city as of 2025 to get a concrete sense of where urban greenery is concentrated. The data includes five files: `listings.csv`, `reviews.csv`, `calendar.csv`, `neighbourhoods.csv`, `neighbourhoods.geojson`, and an urban-forest tree inventory.

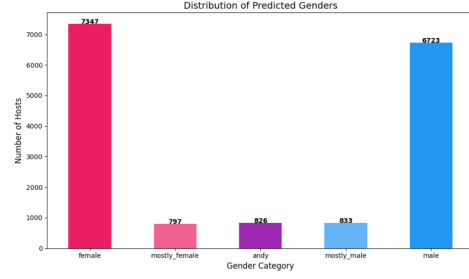


Figure 2: Number of Female identified hosts (left, red) to Male identified hosts (right, blue).

There are 13860 unique host IDs. Reviews-per-month and yearly availability both have heavy spikes at zero [3]. The spike on availability at 365 days is a giveaway that a large group of listings are professionally managed and permanently bookable, while the zero-spike represents listings that are essentially inactive on the platform.

2 Exploratory Data Analysis

We explored the listings, reviews, and calendar datasets from Airbnb. The listings dataset contains 24,491 observations corresponding to Airbnb properties located in Melbourne, Australia. This distribution in [1] indicates that most hosts manage only a single property, suggesting that the majority of listings are offered by individual or occasional hosts. A small group of professional hosts manages a large portfolio of properties. In particular, 10 hosts are responsible for more than 70 listings each, highlighting the presence of highly commercialized operators in the Melbourne Airbnb market. We also ran first names through a gender guesser to get a rough sense of who is hosting [2].

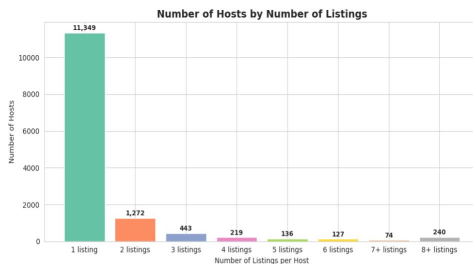


Figure 1: Host listing distribution

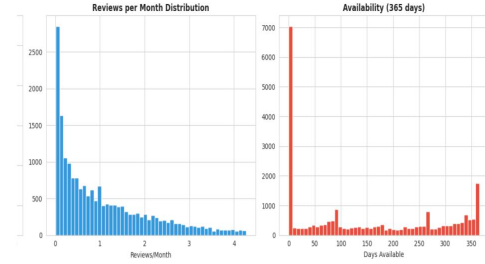


Figure 3: Distributions of reviews-per-month (left) and availability over the next 365 days (right).

In 4, we have shown the geographic concentration. Not to mention, Yarra and Port Phillip are also among the park rich inner cities.

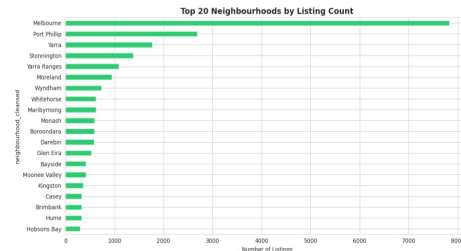


Figure 4: Top 20 neighbourhoods by listing count

3 Natural Language Processing tools

We wrote a regex per amenity and count matches showing Parking is the most mentioned amenity by far [5].

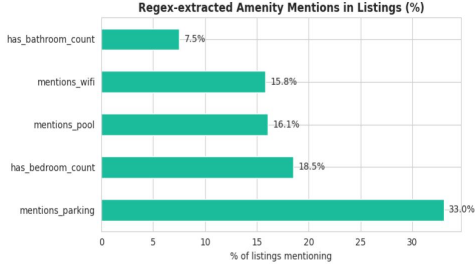


Figure 5: Share of listings whose free-text description contains each amenity keyword

We also found out that bigrams and trigrams over the listing descriptions surface what hosts choose to emphasise [6]. Tf-IDF on this data showed that strongest co-occurrences sit between *melbourne*, *apartment*, *city*, *cbd*, *walk*, *located and street* [7].

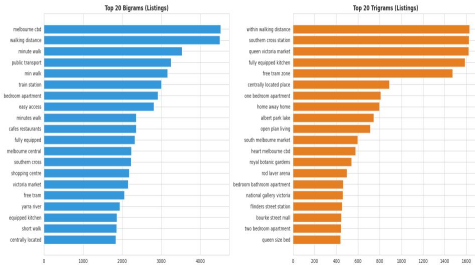


Figure 6: Top 20 bigrams and trigrams in listing descriptions

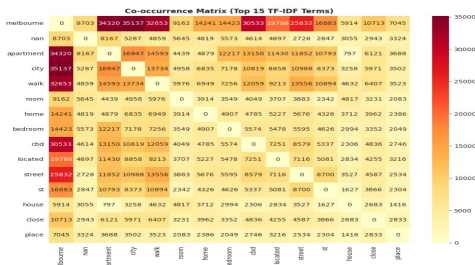


Figure 7: Co-occurrence counts among the top-15 TF-IDF terms

Further analysis on NLP tools have been shown in the appendix.

4 Sentiment analysis

Three complementary lenses are used: the full set of linguistic and technical tools deployed across the pipeline, aspect-level sentiment decomposition and a targeted green-space sentiment test that directly addresses the research question.

4.1 Technical Features and Analytical Framework

We analyse the review corpus using NLP methods applied sequentially. Sentiment scoring uses the VADER (Valence Aware Dictionary and sEntiment Reasoner) compound score, which maps each review to $[-1, +1]$. Text clustering was performed using TF-IDF vectorisation (300 features) followed by k -means, with the optimal number of clusters ($k = 10$) selected via silhouette analysis. Word2Vec embeddings (vector size 100, window 5, minimum count 3) were trained on the first 2,000 tokenised reviews and projected to two dimensions with t-SNE. In the embedding space, green-space terms (*park*, *garden*, *view*, *tree*) cluster consistently near positively-valenced descriptors (*comfortable*, *quiet*), providing an unsupervised confirmation of the sentiment signal.

Part-of-speech tagging revealed that the 20 most frequent adjective-noun pairs include *central-location*, *leafy-neighbourhood*, *beautiful-park*, and *quiet-street*, all of which align with the Location and Views aspects identified in the aspect-based layer. Affect scoring assigned Joy as the dominant emotion class across the corpus (See Figure 8), while Discrepancy dominated the cognition dimension—reflecting the hedging language (*but*, *however*, *should*) reviewers use to qualify otherwise positive stays.

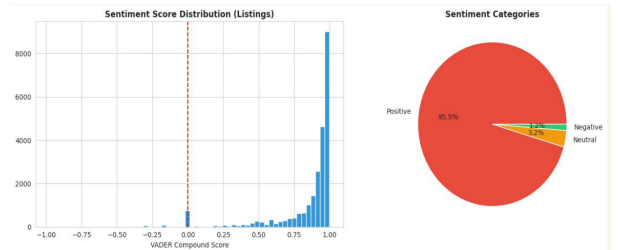


Figure 8: 96% of reviews are positive · Mean VADER score: 0.802 so guests largely satisfied.

4.2 Aspect-Based Sentiment Analysis

Eight aspect lexicons were scanned across every review. They were Cleanliness, Location, Comfort, Amenities, Host, Value, Views and Noise. Whenever a review matched a keyword for a given aspect, its VADER compound score was attributed to that aspect’s tally. Figure 9 shows the mean sentiment and mention frequency by aspect.

**Mean VADER Sentiment by Aspect
(selected values)**

Aspect	Mean VADER Score
Host	+0.810
Views	+0.770
Location	+0.760
Noise	+0.750
Cleanliness	+0.680
Comfort	+0.590
Value	+0.540
Amenities	+0.460

Figure 9: Mean VADER compound sentiment per aspect.

Host quality is rated highest ($\mu = +0.810$), reflecting the well-documented importance of host responsiveness on sharing platforms. Crucially for this study, Views is the most-mentioned aspect ($n = 2,904$ occurrences), making it both the most salient topic and a direct green-space proxy. Location ranks third in mention frequency while maintaining a strong mean sentiment of +0.760. Together, Views and Location appear in approximately three of every four reviews in the corpus. Named Entity Recognition reinforces this picture: geographic place names (GPE, 23.3% of entities) and facilities such as parks and gardens (FAC, 76.7%) dominate the entity type distribution, with parks and gardens constituting the most frequent FAC entities overall.

LDA topic modelling, selecting an optimal $k = 3$ by coherence (u_mass), isolates outdoor/-nature as one of three latent topics (See Figure 17). This unsupervised result converges with the supervised aspect analysis: nature-related language is not peripheral. It forms a coherent cluster of co-occurring terms that reviewers deploy when describing their stay.

4.3 Green-Space Sentiment Analysis

Reviews are labelled *green-mentioning* if they contain at least one term from the set {park, garden, green, nature, tree, forest, trail, botanical, reserve, lake, river, creek, leafy, view}, and their VADER compound scores are compared with those of the remaining reviews.

At the listing description level, 86.4% of listings include at least one green-space keyword. The most frequent terms are *walk* ($\approx 16,000$ occurrences), *park* ($\approx 7,000$), *walking* ($\approx 6,100$), *views* ($\approx 5,900$), and *beach* ($\approx 5,100$), indicating that outdoor access is positioned as a primary selling point by hosts. The green keyword score distribution is right-skewed, with most listings scoring between 1 and 5, and a long tail extending beyond 15.

At the review level, green-mentioning reviews ($n = 3,461$, $\mu = +0.745$) score 0.189 sentiment points higher than non-green reviews ($n = 1,539$, $\mu = +0.556$).

Listings that mention green spaces receive on average 93% more reviews than those that do not (mean review count: 41.3 vs. 21.4), using review count as a demand proxy. It is shown on Figure 16. At neighbourhood level, a positive relationship is visible between mean green score and mean review count, with the highest-demand neighbourhoods clustering above a green score of 4.5. Green language is geographically concentrated in Port Phillip, Yarra, and bayside neighbourhoods, (See Figure 11) consistent with their proximity to parks, the Yarra River, and coastal open space (See Figure 10).

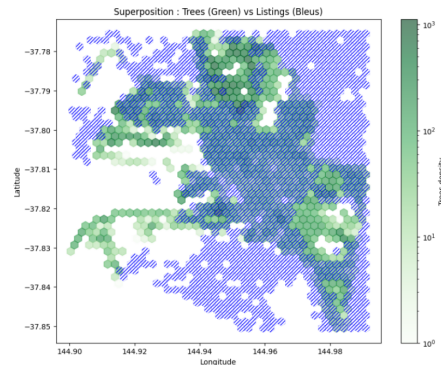


Figure 10: Superposition of Trees density (Green) and Listings density (Blue lines) in the City of Melbourne

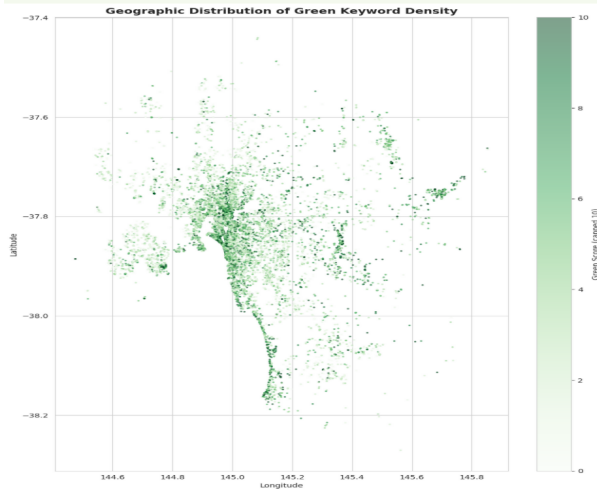


Figure 11: Green language is spatially concentrated in Port Phillip, Yarra, and bayside neighbourhoods.

A Pearson correlation of $r = 0.256$ ($p < 0.01$) is observed between green keyword density (score) and the VADER compound score of individual listing descriptions—a moderate positive relationship. The regression slope of 0.0117 indicates that each additional green keyword is associated with a +0.012 increase in listing-level sentiment.

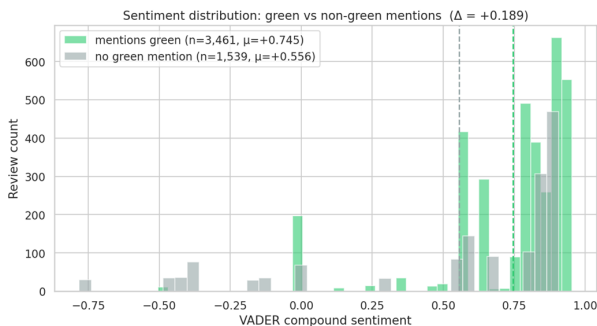


Figure 12: Correlation between linguistic features, and occupancy. Occupancy is strongly tied to review volume (by construction in the occupancy proxy used here).

5 Conclusion

We asked how strongly green spaces in Melbourne neighbourhoods affect Airbnb supply and demand. Evidences seem to converge to

the fact that there is a moderate in strength, and consistent relationship across every analytical method applied, though it remains correlational rather than causal.

On the demand side, reviews that mention green-space vocabulary score 0.189 VADER sentiment points higher than those that do not, and listings featuring green language attract on average 93% more reviews, the study’s primary proxy for demand. The Pearson correlation between green keyword density and listing-level sentiment sits at $r = 0.256$, a moderate but statistically significant signal. It is consistent with other methods as LDA topic modelling independently surfaces outdoor/nature as one of only three latent topics in the corpus. Word2Vec embeddings place terms like *park*, *garden*, and *view* in the neighbourhood of positively-valenced descriptors. Plus, named entity recognition confirms that parks and gardens are the most frequently identified facilities across all reviews.

On the supply side, green language is geographically concentrated in Port Phillip, Yarra, and bayside neighbourhoods, precisely the areas with the highest listing density and the richest tree cover. It suggests that hosts actively position proximity to green space as a selling point, with 86.4% of listing descriptions including at least one green-space keyword.

There is still one question. Guests who have enjoyed a pleasant stay may simply be more disposed to notice and mention a nearby park, rather than the greenery itself generating satisfaction. Nonetheless, the convergence of unsupervised and supervised methods makes it difficult to dismiss the finding as noise. Green spaces do shape how Melbourne’s Airbnb market is perceived and, by extension, how demand is distributed across the city. Thus, it can add evidence to the debate about who benefits from urban ecological investment when short-term rental platforms are part of the housing landscape. Further researches could explore in depth confounding variables such as how green neighbourhoods in Melbourne tend to also be wealthier and closer to the CBD. They could build on a more comprehensive dataset that would include Airbnb prices to compute the green premium.

6 Appendix

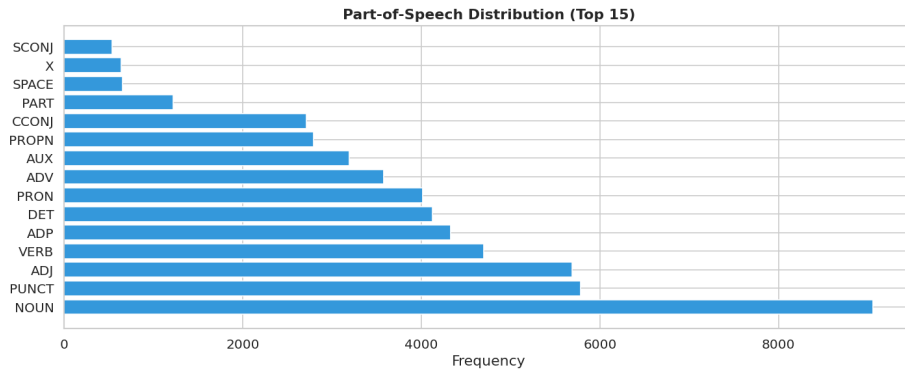


Figure 13: Parts of speech distribution in reviews data

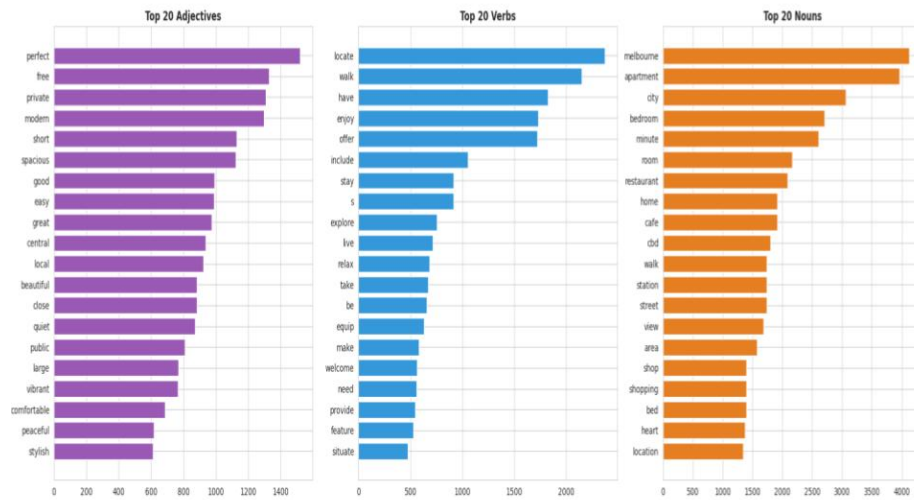


Figure 14: Top 20 adjectives, verbs and nouns in listing descriptions.

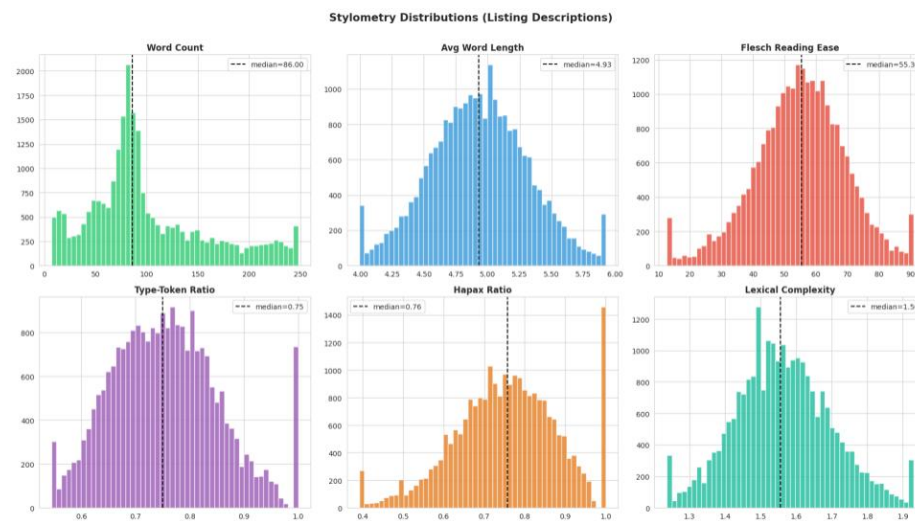


Figure 15: Distributions of stylometric features across listing descriptions

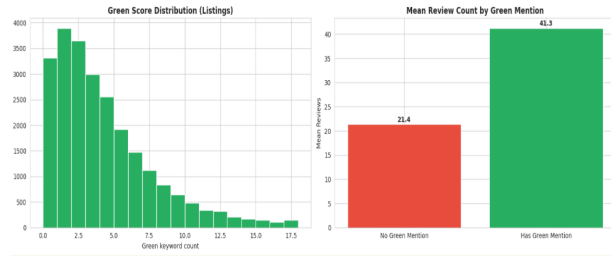


Figure 16: Listings with green mentions receive 93% more reviews on average than those without.

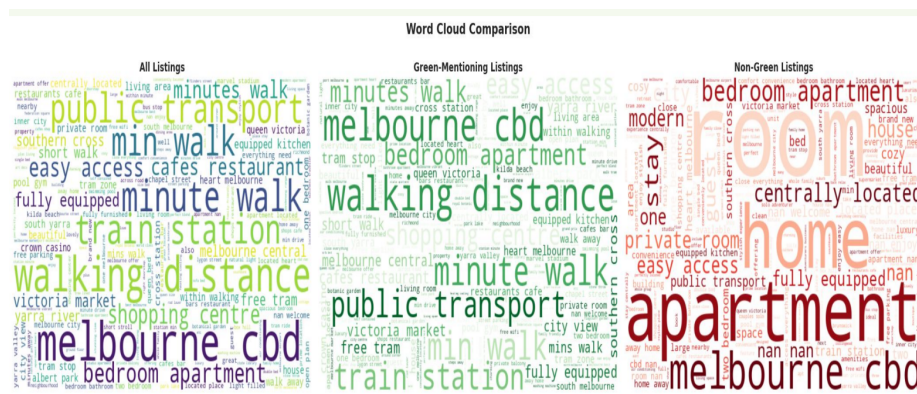


Figure 17: Green listings emphasise outdoor, beach, garden language; non-green listings focus on room-level and urban amenities.